

## CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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## Case 30-2022: A Newborn Girl with Hypoglycemia

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## PRESENTATION OF CASE

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*Dr. Numa P. Perez (Surgery):* A newborn girl was transferred to the neonatal intensive care unit (ICU) of this hospital because of hypoglycemia.

The patient's mother was a 27-year-old woman (gravida 1, para 1) who had received prenatal care at another hospital. At 34 weeks of gestation, an ultrasonographic survey of the fetal anatomy had reportedly revealed an avascular intraabdominal cystic mass that was located near the bladder and measured 3.7 cm by 2.1 cm by 2.0 cm. A sacral dimple was noted. The patient's mother was referred to the maternal–fetal medicine clinic of the other hospital; she declined follow-up imaging and amniocentesis.

At 37 weeks 6 days of gestation, premature labor occurred, and the patient's mother was taken to a second hospital. A fetal heart tracing was noted to be non-reactive with minimal variability. The fetal biophysical profile score was 4 out of 8, indicating decreased breathing motion and body movement but normal tone and amniotic fluid volume. An emergency cesarean section was performed.

On delivery, the newborn had limp tone and poor respiratory effort. The birth weight was 2963 g (19th percentile), the length 48 cm (20th percentile), and the head circumference 35 cm (77th percentile). The Apgar scores at 1 minute and 5 minutes were 7 and 9, respectively. The patient was transferred to the newborn nursery.

In the nursery, the patient was noted to be jittery. Oxygen therapy was administered through a low-flow nasal cannula. The blood glucose level was reportedly less than 10 mg per deciliter (0.6 mmol per liter; reference range, 70 to 110 mg per deciliter [3.9 to 6.1 mmol per liter]) and the bilirubin level 16.0 mg per deciliter (274  $\mu$ mol per liter; reference range, 2.0 to 15.0 mg per deciliter [34 to 257  $\mu$ mol per liter]). A chest radiograph showed a normal cardiac shadow and findings indicative of retained fetal lung fluid. The newborn was fed, and a bolus of dextrose was administered; however, a repeat blood glucose level was 2 mg per deciliter (0.1 mmol per liter). A continuous infusion of dextrose was started. Phototherapy and empirical treatment with ampicillin and gentamicin were administered.

During the next 3 days, the patient's blood glucose level remained low, with a maximum level of 61 mg per deciliter (3.4 mmol per liter), even though the glucose infusion rate was increased and enteral feeding with formula was initiated. On the fourth day of life, she was transferred to the neonatal ICU of this hospital for further treatment.

On admission to the neonatal ICU, additional history was obtained. Before the ultrasonographic survey at 34 weeks of gestation, the pregnancy had been normal. The patient's mother had taken prenatal vitamins. Prenatal maternal screening had been negative for gonorrhea, chlamydia, and syphilis, as well as human immunodeficiency virus, hepatitis B virus, and group B streptococcal infections; there was evidence of immunity to rubella virus. The maternal ABO blood type was AB, Rh positive. The newborn's parents had immigrated to the United States from the Caribbean and now lived in a suburb in New England. The patient's mother, father, and half brother were healthy.

On examination, the patient was alert and appeared comfortable. The temperature was 36.9°C, the heart rate 148 beats per minute, the blood pressure 79/45 mm Hg, the respiratory rate 49 breaths per minute, and the oxygen saturation 98% while she was breathing ambient air. The anterior and posterior fontanelles were flat and soft. The lungs were clear on auscultation. The abdomen was soft without distention, tenderness, or palpable masses. Two sacral dimples with visible bases were noted. The female genitalia appeared normal, but the anus was displaced anteriorly.

The blood glucose level was 60 mg per deciliter (3.3 mmol per liter) while the patient was receiving dextrose at a glucose infusion rate of 10.5 mg per kilogram of body weight per minute. The total bilirubin level was 20.4 mg per deciliter (349  $\mu$ mol per liter). Other laboratory test results are shown in Table 1.

*Dr. Sjik J. Westra:* A chest radiograph showed clear lungs. Abdominal and pelvic ultrasonography (Fig. 1) revealed a large solid–cystic mass anterior to the sacrum that measured 2.7 cm by 3.0 cm by 5.7 cm, with the rectum passing anteriorly to the mass. Results of spinal ultrasonography were normal, as were results of cranial ultrasonography performed through the anterior fontanelle. In addition, magnetic resonance im-

**Table 1. Laboratory Data.\***

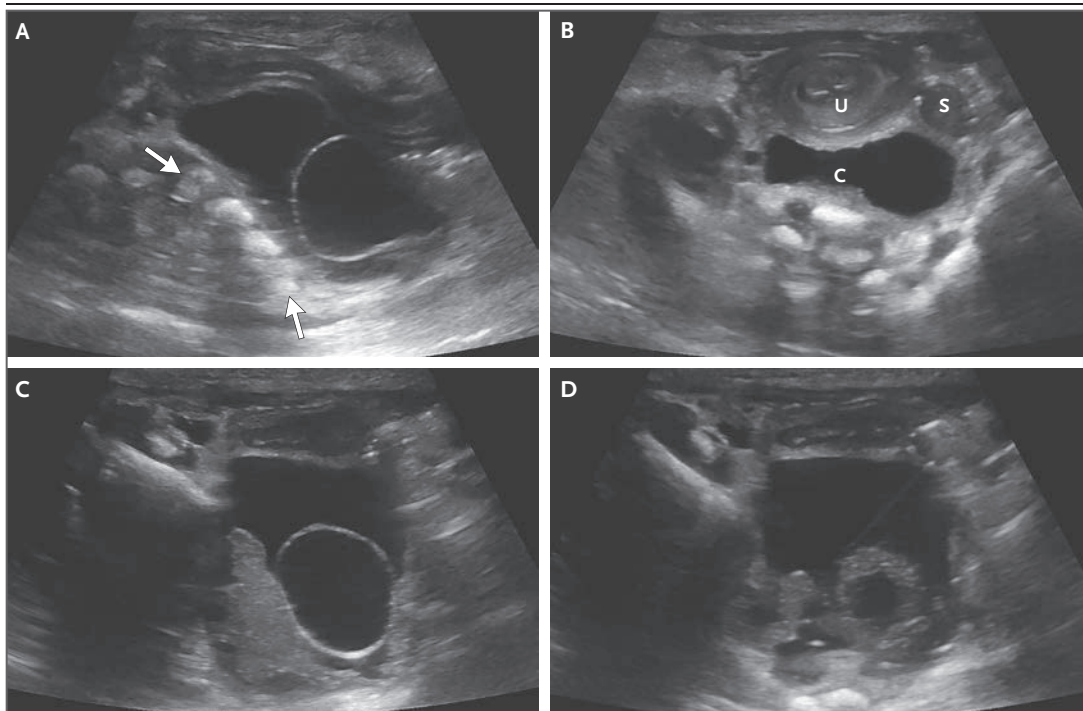
| Variable                             | Reference Range, Children† | On Admission |
|--------------------------------------|----------------------------|--------------|
| Hemoglobin (g/dl)                    | 14.5–22.5                  | 17.2         |
| Hematocrit (%)                       | 45.0–67.0                  | 51.2         |
| Platelet count (per $\mu$ l)         | 150,000–400,000            | 95,000       |
| White-cell count (per $\mu$ l)       | 9400–34,000                | 8630         |
| Reticulocytes (%)                    | 2.1–4.8                    | 7.0          |
| Haptoglobin (mg/dl)                  | 30–200                     | <10          |
| Sodium (mmol/liter)                  | 135–145                    | 142          |
| Potassium (mmol/liter)               | 3.4–5.0                    | 3.7          |
| Chloride (mmol/liter)                | 98–106                     | 109          |
| Carbon dioxide (mmol/liter)          | 22–27                      | 21           |
| Urea nitrogen (mg/dl)                | 5–20                       | 6            |
| Creatinine (mg/dl)                   | 0.30–1.00                  | 0.77         |
| Glucose (mg/dl)                      | 70–110                     | 60           |
| Calcium (mg/dl)                      | 8.5–10.5                   | 9.3          |
| Uric acid (mg/dl)                    | 2.3–6.6                    | 4.0          |
| Lactate dehydrogenase (U/liter)      | 110–210                    | 1173         |
| Albumin (g/dl)                       | 3.3–5.0                    | 3.2          |
| Globulin (g/dl)                      | 1.9–4.1                    | 1.1          |
| Total protein (g/dl)                 | 6.0–8.3                    | 4.3          |
| Direct bilirubin (mg/dl)             | 0.0–0.4                    | 2.2          |
| Total bilirubin (mg/dl)              | 2.0–15.0                   | 20.4         |
| Alanine aminotransferase (U/liter)   | 7–33                       | 10           |
| Aspartate aminotransferase (U/liter) | 45–150                     | 37           |
| Alkaline phosphatase (U/liter)       | 83–248                     | 216          |

\* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for uric acid to micromoles per liter, multiply by 59.48. To convert the values for bilirubin to micromoles per liter, multiply by 17.1.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. They may therefore not be appropriate for all patients.

aging (MRI) of the abdomen, pelvis, and spine (Fig. 2) showed a presacral solid–cystic mass that measured 2.7 cm by 2.8 cm by 5.9 cm, with a small extension into the ischiorectal fossa on the right side. There was no evidence of abdominal lymphadenopathy or involvement of the sacral spine.

*Dr. Perez:* During the next few days, the patient continued to receive dextrose at a high glucose infusion rate to maintain normoglycemia. At 5 days of life, a small bowel movement occurred. The



**Figure 1. Ultrasound Images of the Pelvis.**

A midsagittal image (Panel A) shows a complex cystic mass (arrows) anterior to the sacrum. A transverse image obtained through the superior aspect of the mass (Panel B) shows the relationship of the cystic mass (C) with the uterus (U) and the sigmoid colon (S). Consecutive transverse images obtained more inferiorly (Panels C and D) show that the mass has a complex nature, with solid components at the right margin.

blood lactate dehydrogenase level was 1173 U per liter (reference range, 110 to 210) and the beta human chorionic gonadotropin level 10.7 mIU per milliliter (reference range, <1.1).

A diagnostic test was performed.

#### DIFFERENTIAL DIAGNOSIS

*Dr. Sarbattama Sen:* Blood glucose levels are physiologically low immediately after birth and approach levels observed in older children or adults after 72 hours of life.<sup>1</sup> This newborn girl had persistent hypoglycemia postnatally, for which she received a continuous infusion of dextrose to maintain normoglycemia. To determine the cause of her prolonged hypoglycemia, it is helpful to review the physiology of postnatal glycemic adaptation.

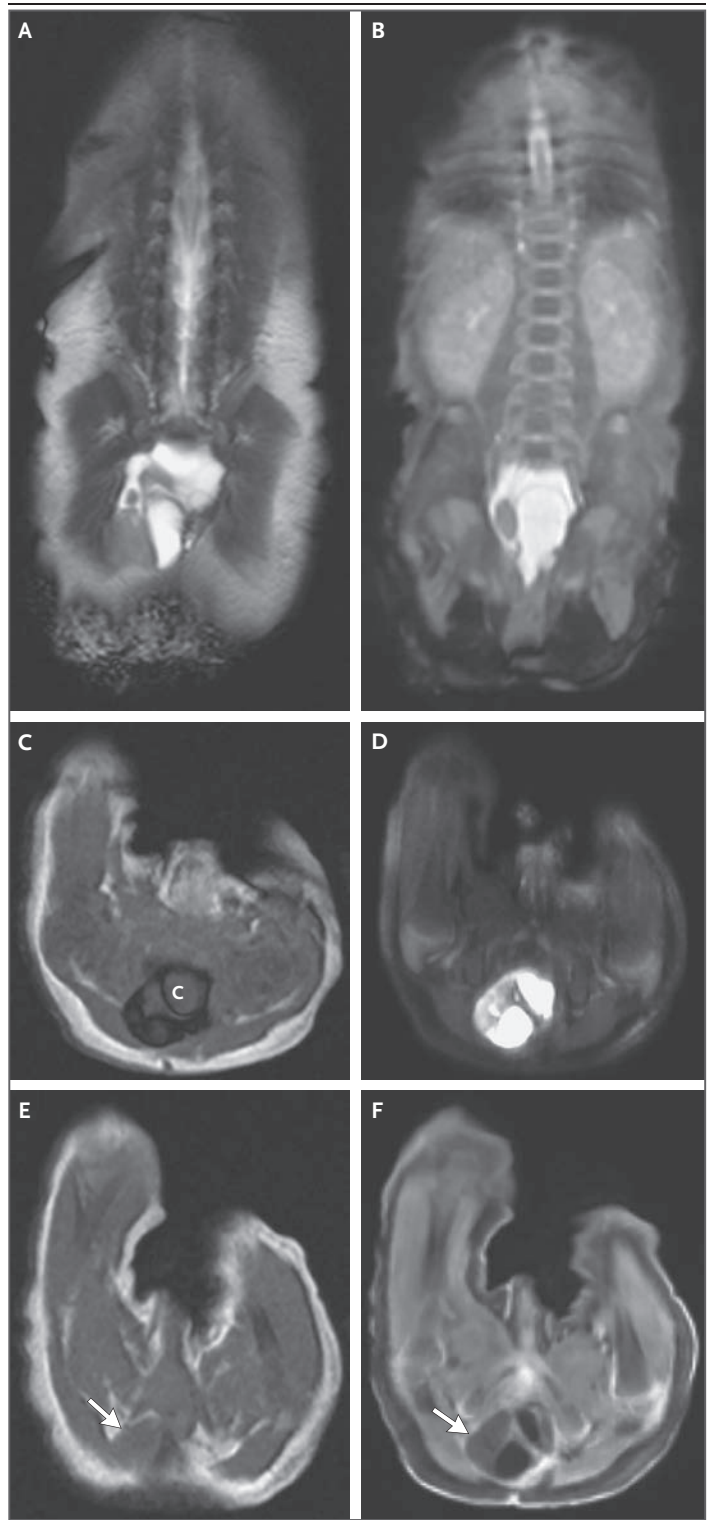
#### PHYSIOLOGY OF POSTNATAL GLYCEMIC ADAPTATION

The primary source of energy for the fetus is glucose, which the mother provides to the fetus through a concentration gradient.<sup>2-4</sup> The fetal

glucose level is approximately 80% of the maternal level. The glucose level at which insulin is secreted is lower in a fetus than in a newborn, given that the function of insulin in utero is primarily related to fetal growth rather than glucose regulation.<sup>5</sup> Before birth, the maternal and fetal glucose levels increase until clamping of the umbilical cord, when the maternal glucose supply to the infant is interrupted. Thereafter, the neonatal glucose level decreases, reaching a nadir between 1 and 3 hours after birth.<sup>6</sup> In response to the falling glucose level, the secretion of insulin typically decreases while the secretion of glucagon and catecholamines increases, stimulating glucose production through gluconeogenesis and glycogenolysis. The catecholamine surge associated with hypoglycemia increases the production of lactate, which serves as an alternative fuel for gluconeogenesis.<sup>6</sup> Ultimately, the glucose level at which insulin is secreted by pancreatic beta cells typically adjusts to the higher, adult threshold,<sup>1</sup> allowing the infant to maintain normoglycemia.<sup>7,8</sup>

**Figure 2. MRI of the Pelvis.**

Consecutive coronal T2-weighted, fluid-sensitive, fast spin-echo images (Panels A and B) show a complex, fluid-filled pelvic mass that is not continuous with the thecal sac. Correlative axial T1-weighted and T2-weighted images (Panels C and D, respectively) show the complex nature of the mass, with mixed fluid-filled and solid components; a cystic component (C) is associated with an intermediate T1 signal, which is indicative of a high level of proteinaceous content. Correlative axial T1-weighted images obtained without fat suppression before the administration of gadolinium and with fat suppression after the administration of gadolinium (Panels E and F, respectively) show a poorly enhancing, solid extension of the mass (arrows) into the ischio-rectal fossa on the right side.

**TYPICAL RISK FACTORS FOR NEONATAL HYPOGLYCEMIA**

There are maternal, fetal, and neonatal conditions that can confer a predisposition for delayed or exaggerated glycemic transition in an infant and potentially lead to neonatal hypoglycemia. These conditions are associated with three mechanisms: decreased insulin sensitivity, decreased stores of glycogen, and increased insulin secretion. Infants who are born preterm or are small for gestational age may be affected by any or all of these mechanisms.<sup>9-11</sup> Infants who are large for gestational age are thought to be affected by increased insulin secretion in utero that resulted in fetal overgrowth, regardless of whether the mother had received a diagnosis of diabetes.<sup>12</sup> When a mother has diabetes, fetal hyperinsulinism develops in response to maternal hyperglycemia.<sup>13</sup> In this infant, none of these typical prenatal or perinatal risk factors for hypoglycemia were identified.

In the absence of these typical risk factors, other causes of severe, prolonged hypoglycemia should be considered. These include foci of ectopic or dysregulated insulin production (congenital hyperinsulinism or another focus of ectopic insulin production), metabolic diseases (fatty-acid oxidation defects, organic acidemias, urea-cycle defects, or glycogen-storage disease), genetic syndromes (the Prader-Willi syndrome, the Beckwith-Wiedemann syndrome, the Kabuki syndrome, the Sotos syndrome, or the Turner syndrome), and endocrinopathies (hypopituitarism, growth-hormone deficiency, or congenital adrenal hyperplasia).

**DIAGNOSTIC FASTING STUDY**

A diagnostic fasting study could determine whether the hypoglycemia is due to hyperinsulinism.<sup>14</sup> During the study, all exogenous (enteral

and parenteral) sources of glucose are held. Once the blood glucose level is less than 50 mg per deciliter (2.8 mmol per liter), the following levels are obtained: insulin, free fatty acids,  $\beta$ -hydroxybutyrate, ammonia, lactate, cortisol, growth hormone, acylcarnitines, and urine organic acids. In the context of hyperinsulinism, the insulin level is inappropriately high, whereas the insulin level is typically suppressed under conditions of hypoglycemia. Inappropriately high insulin secretion leads to suppression of the lipase level, which results in low levels of free fatty acids and  $\beta$ -hydroxybutyrate and also inhibits amino acid breakdown, thus resulting in a low ammonia level. The lactate level may be high, given the ongoing catecholamine response. A diagnostic fasting study also provides information about central endocrinopathies, such as hypopituitarism (identified by detection of a low cortisol level), growth-hormone deficiency, fatty-acid oxidation defects (identified by assessment of the acylcarnitine profile), and urea-cycle defects (identified by assessment of the urine organic acid profile).

#### HYPERINSULINISM AND SACROCOCYGEAL TERATOMA

If a diagnostic fasting study confirms the presence of hyperinsulinism, the next question is whether the hyperinsulinism is related to the other prominent clinical finding in this case: a cystic mass in the sacral region. By far, the most common pathologic masses that occur in this anatomical location and with this radiographic appearance are sacrococcygeal teratomas, which consist of heterogeneous, pleiotropic cells.<sup>15,16</sup> In a patient with a sacrococcygeal teratoma, hyperinsulinism can be caused by a syndrome that has led to the sacrococcygeal teratoma and hyperinsulinism or by a focus of inappropriate insulin secretion within the teratoma (or both).

The Sotos syndrome is a rare disorder that is associated with hyperinsulinism and sacrococcygeal teratomas. Newborns with the Sotos syndrome can present with few physical manifestations.<sup>17</sup> The Sotos syndrome is identified by the detection of pathogenic variants in the gene *NSD1*; more than 380 variants in this gene have been linked to this disease.<sup>18</sup>

Trisomy 13 has been associated with sacrococcygeal teratomas and hypoglycemia in rare cases,

but in this context, hypoglycemia is less likely to be related to hyperinsulinism than to growth restriction and poor glycogen stores. Newborns with trisomy 13 typically present with a constellation of physical manifestations, which were not described in this patient.<sup>19</sup>

A focus of ectopic insulin production within the sacrococcygeal teratoma can be identified by means of abdominal and pelvic positron-emission tomography (PET) with <sup>18</sup>F-fluorodihydroxyphenylalanine.<sup>20</sup> Alternatively, if surgical resection of the mass is planned, pathological examination of the resection specimen can include an evaluation for islet cells.

On the basis of the patient's presentation with persistent hypoglycemia in the context of a probable sacrococcygeal teratoma, I suspect that the Sotos syndrome is the most likely diagnosis. To establish this diagnosis, targeted sequencing should be performed to identify variants in *NSD1*. Alternatively, whole-exome sequencing could be performed in the infant, as well as in the parents if possible (trio approach), to identify pathogenic variants associated with genetic disorders that could be consistent with the patient's clinical presentation, including the Sotos syndrome.

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#### DR. SARBATTAMA SEN'S DIAGNOSIS

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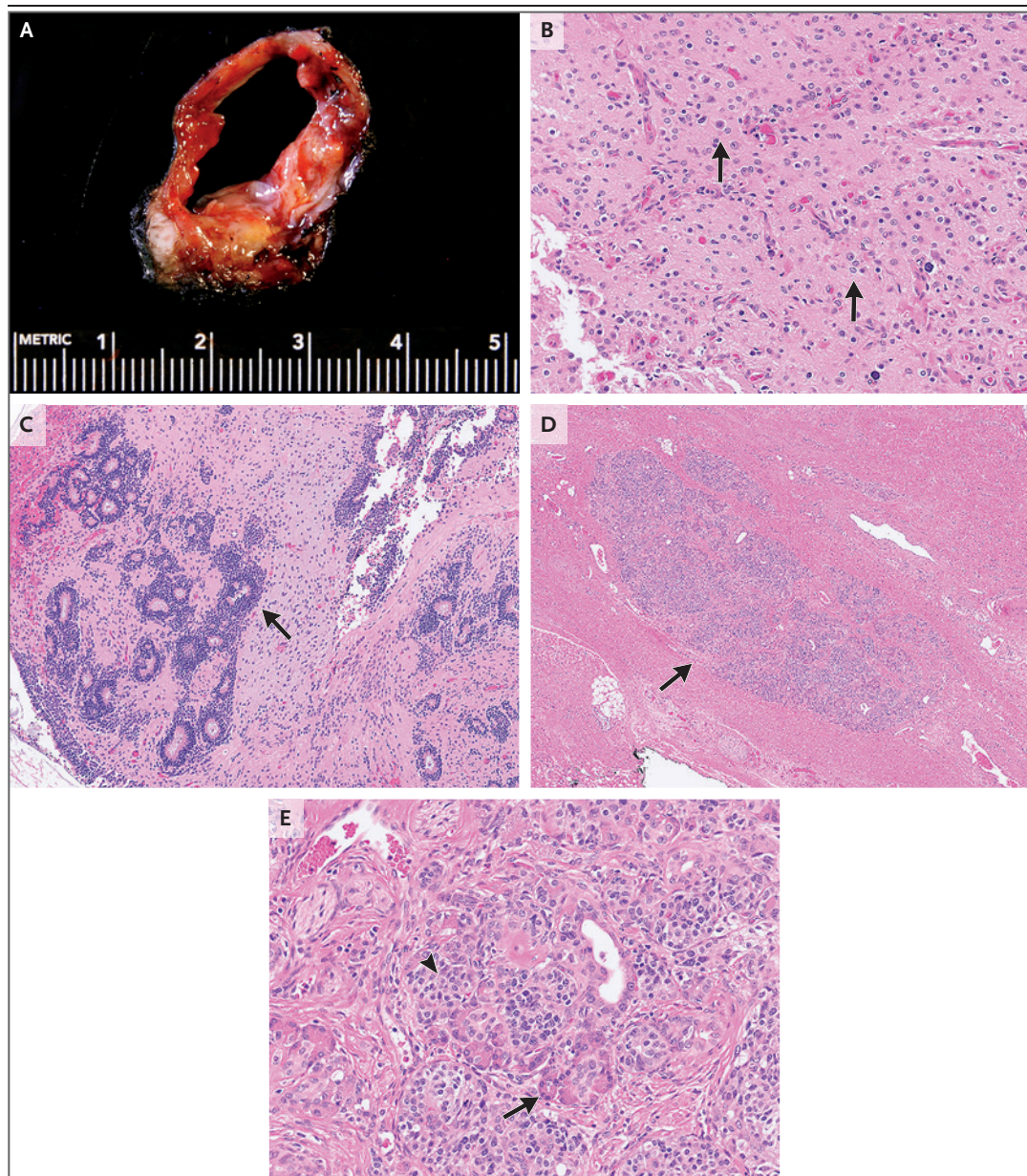
Hyperinsulinism related to ectopic insulin production from a sacrococcygeal teratoma, with possible underlying Sotos syndrome.

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#### PATHOLOGICAL DISCUSSION

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*Dr. Melanie C. Kwan:* The first diagnostic approach in this case was surgical resection of the sacral mass. On gross examination, the mass measured 4.3 cm by 3.3 cm by 2.7 cm. It appeared intact and had a pink-purple outer surface. Sectioning of the mass showed a cystic structure that contained clear, thick fluid (Fig. 3A). Histologic examination revealed predominantly mature neural tissue, along with small areas of immature neural tissue (Fig. 3B and 3C). Other elements included respiratory epithelium. There was a single focus of pancreatic acinar and endocrine tissue that contained insulin- and glucagon-secreting cells (Fig. 3D and 3E), which were identified on immunohistochemical staining. Within the focus of pancreatic tissue, ductuloin-



**Figure 3. Resection Specimen.**

On gross examination, the resected mass has a cystic structure, and a cyst wall has areas of tan-yellow tissue (Panel A). On hematoxylin and eosin staining, the mass has predominantly mature neural tissue (Panel B, arrows), along with small areas of immature neural tissue (Panel C, arrow). Within the mass, there is a small focus of pancreatic tissue (Panel D, arrow) with endocrine components (Panel E, arrowhead) and exocrine components (Panel E, arrow).

sular complexes were present. The focus constituted less than 5% of the mass. Taken together, these findings were consistent with a sacrococcygeal teratoma.

Teratomas are the most common type of perinatal tumor, comprising approximately 37 to 52% of congenital neoplasms. Teratomas are more

common in female patients than in male patients.<sup>21</sup> Congenital teratomas are most likely to occur in the sacrococcygeal area.<sup>21</sup> Sacrococcygeal teratomas are a type of extragonadal germ-cell tumor, and they typically contain immature neuroepithelium.<sup>22</sup> Unlike with gonadal teratomas, the presence of immature neural tissue in

an extragonadal germ-cell tumor is not predictive of malignant behavior.<sup>22</sup> Extragonadal germ-cell tumors are associated with a better prognosis in pediatric patients, particularly those younger than 12 years of age, than in adults.<sup>22</sup> Somatic malignant components were not found in this case. Evidence of a non-germ-cell cancer in a sacrococcygeal germ-cell tumor is a feature associated with poor prognosis.<sup>22</sup> Extragonadal germ-cell tumors in pediatric patients are cytogenetically distinct from malignant germ-cell tumors in adults, which have the isochromosome 12p abnormality.<sup>22</sup>

#### PATHOLOGICAL DIAGNOSIS

Sacrococcygeal teratoma with neural and pancreatic elements.

#### DISCUSSION OF MANAGEMENT

*Dr. Juan D. Matute:* Current guidelines for the treatment of neonatal hypoglycemia<sup>23,24</sup> were reviewed with several neonatologists and pediatric endocrinologists at this hospital. In accordance with the institutional protocol, we aim to establish a glucose level higher than 45 mg per deciliter (2.5 mmol per liter) within the first 48 hours of life and to establish a level higher than 60 mg per deciliter (3.3 mmol per liter) after 48 hours of life. When neonatal hypoglycemia is due to hyperinsulinism, our goal is to maintain a glucose level higher than 70 mg per deciliter (3.9 mmol per liter), because hyperinsulinism limits the generation of ketones, an alternative energy source for the neonatal brain.<sup>25</sup>

As compared with adults, newborns have an increased brain size relative to body mass; for this reason, the glucose utilization rate in newborns (4 to 6 mg per kilogram per minute) is 2 to 3 times as high as the rate in adults.<sup>24</sup> Glycogenolysis and gluconeogenesis usually result in sufficient glucose production after birth until oral intake is established. However, newborns with impaired endogenous glucose production, excess insulin production, or both might receive glucose from an exogenous source. Enteral or parenteral glucose can be administered to maintain a normal glucose level for the patient's age. Enteral feeding can be offered, or a gel that contains 40% dextrose can be administered to the buccal mucosa. However, it is not uncommon for newborns with hypoglycemia to receive

parenteral glucose. In addition to the administration of glucose, medications such as glucagon, diazoxide, and glucocorticoids may be used to elevate the blood glucose level, with the choice of treatment depending on the underlying mechanism of hypoglycemia.<sup>26</sup>

*Dr. Jordan S. Sherwood:* In this patient, hypoglycemia persisted after resection of the sacrococcygeal teratoma. On laboratory testing in the context of hypoglycemia, levels of ammonia, lactate, cortisol, and growth hormone were normal, as were the acylcarnitine profile and the urine organic acid profile. There was a low level of  $\beta$ -hydroxybutyrate, as well as a low but detectable level of insulin. A detectable level of insulin is inappropriate in the context of hypoglycemia, given that the insulin level is typically suppressed under conditions of hypoglycemia. The detectable insulin level in this patient was consistent with a diagnosis of hyperinsulinism.

For infants with hyperinsulinism who receive dextrose at a persistently elevated glucose infusion rate to maintain an adequate blood glucose level and prevent hypoglycemia, first-line treatment is a trial of diazoxide. Diazoxide increases the blood glucose level by inhibiting the release of insulin through its action on the ATP-sensitive potassium channels that are expressed on pancreatic beta cells.<sup>27</sup> Diazoxide maintains the channels in an open state, which results in membrane hyperpolarization and thereby decreases insulin secretion. An adequate response to diazoxide is defined as maintaining a plasma glucose level higher than 70 mg per deciliter during a monitored fast (i.e., safety fast).<sup>24</sup> The recommended target glucose level is higher for patients with hyperinsulinism than for patients with other forms of neonatal hypoglycemia. In the hyperinsulinemic state, ketogenesis is suppressed, and there is no alternative fuel source for brain metabolism in the context of hypoglycemia. Infants with persistent hyperinsulinism are at risk for poor neurocognitive outcomes, including epilepsy, speech delay, learning disabilities, and behavior problems; therefore, maintaining an adequate glucose level is essential to prevent recurrent hypoglycemia, which can have a negative effect on neurodevelopment.<sup>28</sup>

Infants who do not have a response to diazoxide should be assessed for monogenic causes of hyperinsulinism, because specific monogenic

causes are associated with focal beta-cell hyperplasia, which is treated with surgery. In addition, some forms of diffuse hyperinsulinism, hyperinsulinism due to perinatal stress, and other hyperinsulinism-associated syndromes may be unresponsive to diazoxide. Specialized beta-cell imaging such as PET with  $^{18}\text{F}$ -fluorodihydroxyphenylalanine should be reserved for patients with suspected focal lesions and is often performed only at specialized centers familiar with the treatment of infants with hyperinsulinism.

Diazoxide has several potential adverse side effects, including pulmonary hypertension, fluid retention, neutropenia, thrombocytopenia, and hyperuricemia.<sup>29</sup> More recently, a possible association with necrotizing enterocolitis has been reported.<sup>30,31</sup> Given the risk of pulmonary hypertension and edema, a baseline echocardiogram is typically obtained to screen for underlying cardiac conditions. In addition, a thiazide diuretic is often administered at the same time as diazoxide to help prevent fluid retention. The complete blood count and uric acid level are obtained at baseline and monitored over time. In this patient, the baseline echocardiogram, complete blood count, and uric acid level were normal. Therapy with diazoxide and chlorothiazide was initiated. During a safety fast, the blood glucose level remained higher than 70 mg per deciliter. The parents received instruction regarding glucometer use for home monitoring of blood glucose levels with adjustment of the diazoxide dose as needed to maintain the target blood glucose level. In light of this patient's clinical presentation, genetic testing was performed.

#### GENETIC TESTING

*Dr. Frances A. High:* Genetic testing was performed at the time of admission to the neonatal ICU to identify the cause of the sacrococcygeal teratoma. A chromosome microarray analysis and *MNX1* sequencing to rule out the Currarino syndrome were nondiagnostic. The Currarino syndrome is associated with multiple congenital anomalies, often including sacrococcygeal teratoma. Although this syndrome is characterized by a classic triad that includes other sacral and anorectal anomalies, and these features were absent in this patient, molecular genetic testing was still performed because the clinical spec-

trum of Currarino syndrome is known to be broad.

When the patient was 4 months of age, she was seen in the outpatient genetics clinic. She was still receiving diazoxide treatment for hyperinsulinism. Additional features were noted, including progressive macrocephaly, dolichocephaly, hypertonia, and developmental delays. Macrocephaly was associated with a head circumference of 47.2 cm (>99th percentile; z score, +4.99). Facial features were notable for a broad forehead, posteriorly rotated ears, a flat nasal bridge, anteverted nares, and a pointed chin. MRI of the head revealed ventriculomegaly with a slight increase in the size of the lateral and third ventricles, as compared with measurements observed on imaging that had been performed in the neonatal period. Further genetic tests revealed a frameshift variant in *NSD1* (c.1768\_1769del, p.Leu590Ilefs\*), a finding that led to the diagnosis of the Sotos syndrome.

The Sotos syndrome is a genetic condition with cardinal features of overgrowth, developmental delays, and a characteristic facial appearance.<sup>32</sup> Overgrowth may occur prenatally or postnatally and can manifest as macrocephaly and tall stature. Among patients with the Sotos syndrome, neurodevelopmental findings vary but can include developmental delays, intellectual disability, learning disability, or autism spectrum disorder. The facial features seen in this patient are typical of those observed in infants with the Sotos syndrome; the facial features in some patients change over the course of the life span.

Sacroccygeal teratoma has been reported as a rare presenting feature of the Sotos syndrome.<sup>33</sup> Other tumors can also occur with this syndrome, but the overall risk of cancer is low.<sup>32</sup> Infants with the Sotos syndrome are recognized to be at risk for neonatal hypoglycemia due to transient hyperinsulinism. However, in one case series involving seven patients with hyperinsulinism associated with pathogenic *NSD1* variants, three of the patients had persistent hyperinsulinism for which they received treatment beyond infancy.<sup>17</sup>

Both the biologically active teratoma and the underlying Sotos syndrome are likely causes of hyperinsulinism in this patient. Although the teratoma may have been a contributing factor initially, the persistence of hyperinsulinism after

resection of the teratoma is most likely attributable to the diagnosis of the Sotos syndrome.

The patient's parents were counseled about the Sotos syndrome and told that this diagnosis explained their daughter's sacrococcygeal teratoma, hyperinsulinism, and increased head circumference. They were told to anticipate some developmental delay, the severity of which was difficult to predict. The importance of ongoing developmental supports was emphasized. They were also counseled that the risk for future pregnancies was likely to be low, given that the parents showed no clinical signs of the Sotos

syndrome and that most cases of the Sotos syndrome are due to de novo genetic variants. To confirm the risk, however, the parents were offered and are pursuing genetic testing for the NSD1 variant.

#### FINAL DIAGNOSIS

Neonatal hypoglycemia due to biologically active teratoma and the Sotos syndrome.

This case was presented at Pediatric Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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